Notes: Bandiera, Prat, and Valletti (AER, 2009)

Active and Passive Waste in Government Spending: Evidence from a Policy Experiment

Contents

1 Big picture

- **Question.** The paper asks why some public bodies pay much higher prices than others for the same standardized goods. The key question is not only whether there is government waste, but whether the waste is *active* or *passive*.
- **Active waste.** Active waste benefits the public decision maker directly or indirectly. The canonical example is corruption: an official accepts a bribe or another private benefit in exchange for paying a high procurement price.
- **Passive waste.** Passive waste does not benefit the public decision maker. It can arise from low procurement skill, low incentives to search for low prices, poor organization, cumbersome rules, or lack of managerial capacity.
- **Setting.** The authors study purchases of 21 generic goods by 208 Italian public bodies between 2000 and 2005. The goods include items such as laptops, printers, paper, fuel, phone contracts, and office furniture.
- **Policy experiment.** Italy created a national procurement agency, Consip, that negotiated framework agreements for standardized goods. At any point in time, only some goods were available from Consip. Public bodies could often choose whether to buy from Consip or on the open market.
- **Identification idea.** High open-market prices alone cannot tell active from passive waste. The paper uses the decision to buy from Consip when it is available. A passive-waste body should like Consip because it offers an easy low-price option. An active-waste body should dislike Consip because buying from Consip removes opportunities for private benefits.
- **Main reduced-form result.** Public bodies that pay higher prices when Consip is not available are *more* likely to buy from Consip when it is available. This points toward passive waste as the main source of cross-body price differences.
- **Main structural result.** Under the authors' structural assumptions, at least 82 percent of estimated waste is passive. Passive waste accounts for more than half of total waste for at least 83 percent of public bodies.
- **Governance result.** Waste differs strongly by governance type. Ministries and social security institutions pay much more than universities and health authorities, and the difference is mostly passive rather than active.
- **Economic takeaway.** The paper argues that public procurement problems are not only about corruption. Sheer inefficiency, low competence, and poor governance can be quantitatively more important than active rent extraction, at least for standardized goods.

2 Incremental contribution of the paper

- **From measuring waste to diagnosing waste.** Earlier work often documents that governments overpay or that corruption exists. This paper separates the mechanism behind overpayment into active waste and passive waste.

- **A revealed-preference identification contribution.** The key innovation is to use the choice between Consip and the open market. The same high price has opposite implications for the Consip decision depending on whether the source is active or passive waste.
- **A policy-experiment contribution.** Consip agreements switch on and off across goods and time. This creates variation in whether a given public body can use centralized procurement for a given good.
- **A measurement contribution.** The paper uses detailed purchase-level data: unit prices, quantities, brands, models, delivery conditions, and good characteristics. This allows the authors to compare prices for highly standardized goods.
- **A governance contribution.** The paper links procurement performance to institutional form: Napoleonic central bodies, US-style local governments, and semi-autonomous bodies. The results suggest that autonomy and accountability matter for passive waste.
- **A structural decomposition contribution.** The model estimates public-body-level passive and active waste parameters, rather than only reporting reduced-form correlations.
- **A policy-design contribution.** If waste is mostly active, the solution is stricter monitoring and rules. If waste is mostly passive, stricter rules can backfire; better procurement support, skill, discretion, and centralized options may be more useful.
- **A cautionary contribution.** The paper does not deny that corruption matters in Italy. Instead, it shows that the corruption-centered view can miss an even larger source of inefficiency in standardized public procurement.

3 Institutional background and policy experiment

3.1 Types of Italian public bodies

- **Napoleonic bodies.** Ministries and central administration follow a top-down civil-service model. These bodies have less managerial autonomy and are typically controlled by entrenched civil servants.
- **US-style local bodies.** Regions, provinces, and towns have directly elected leaders with broad powers. Local politics tends to focus more on practical local issues and candidate characteristics.
- **Semi-autonomous bodies.** Health authorities and universities have more budgetary and administrative autonomy. Health authorities are headed by directors under private-law contracts and may have performance-related pay.
- The paper uses these differences to ask whether governance structure predicts procurement prices and the type of waste.

Interpretation: The governance classification is central because the paper's results are not mainly driven by geography, size, or social capital proxies. They are strongly related to institutional form.

3.2 Consip framework agreements

- Consip is a national procurement agency created to reduce public expenditure on goods and services.

- Consip negotiates framework agreements with suppliers. Under an active agreement, a supplier commits to sell a specified product at a specified price and condition to any Italian public body.
- Public bodies can order through the Consip catalog, by fax, or by phone.
- Agreements have start dates, end dates, and maximum quantities. At a given time, only a subset of goods in the data is available from Consip.
- The legal obligation to use Consip varies over time and across public bodies. The choice was free for all bodies in 2004 and 2005. In 2003 all bodies were formally required to use Consip for equivalent goods, and central bodies faced stronger requirements earlier.
- Even when Consip purchases were formally mandatory, public bodies could justify outside purchases by claiming that Consip goods did not match their needs.

Interpretation: Consip is not used simply as a price benchmark. It is also an alternative purchase channel that changes the public body's choice set. That is what allows the paper to infer the nature of waste.

4 Data and measurement

4.1 Purchase-level survey data

- The data come from a survey designed and implemented by the Italian Statistical Agency, ISTAT.
- The survey covers purchases between 2000 and 2005 and was administered in three annual rounds between 2003 and 2005.
- The sample focuses on generic goods that are comparable, widely purchased, and relevant for public budgets.
- The final working sample contains 6,068 purchase observations for 21 goods by 208 public bodies.
- For each purchase, the data include price, date, quantity, and detailed characteristics such as brand, model, technical specifications, and delivery conditions.
- On average, 52 percent of purchases are made when a Consip agreement is active and 48 percent when no agreement is active.

Interpretation: The paper's empirical power comes from seeing the same public bodies purchase multiple comparable goods, sometimes when Consip is available and sometimes when it is not.

4.2 Outcome variables

- The main price outcome is the normalized log unit price paid for a purchase.
- Prices are adjusted for month and year using the consumer price index.
- The key open-market price measure is a public-body fixed effect estimated only from purchases made when no Consip agreement is active.

- The key Consip-choice outcome is an indicator equal to one if the public body buys from Consip when a Consip agreement is active.

Interpretation: The fixed effect summarizes whether a public body systematically pays high or low prices for equivalent goods when it cannot use Consip.

4.3 Main descriptive facts

- The public body at the ninetieth percentile of the estimated price distribution pays about 55 percent more than the public body at the tenth percentile.
- If all public bodies paid the same prices as the tenth-percentile body, sample expenditure would fall by about 21 percent. Excluding bodies below the tenth percentile, savings would be about 27 percent.
- Since public purchases of goods and services are roughly 8 percent of GDP, a representative extrapolation would imply potential savings between 1.6 and 2.1 percent of GDP.
- Prices differ systematically by institutional class. Ministries and social security institutions pay the most; semi-autonomous bodies pay the least.

Interpretation: The raw economic magnitude is large. The central empirical task is therefore to explain whether the high prices reflect rent extraction or inefficient procurement.

5 Models and empirical specifications

5.1 Theoretical model without Consip

Let public body i buy good g at time t. The total price is

$$p_{igt} = f_{gt}(b_{igt}, \mu_i),$$

where b_{igt} is the private benefit chosen by the purchasing manager and μ_i is the public body's passive-waste parameter.

The purchasing manager's payoff is

$$\Omega_{igt} = -p_{igt} + \beta_i b_{igt},$$

where β_i is the public body's active-waste propensity.

- A higher μ_i makes procurement less efficient and increases price without benefiting the manager.
- A higher β_i makes private benefits more attractive and increases the optimal amount of active waste.
- When Consip is absent, high prices can be generated either by high passive waste μ_i or high active waste β_i.

Interpretation: Price data alone are observationally equivalent: both corruption-like behavior and incompetence-like behavior show up as higher prices.

5.2 Theoretical model with Consip

When Consip is active, the public body can either buy from Consip or buy outside Consip.

- Consip charges the same price to all public bodies for a given agreement.
- Buying from Consip is assumed to generate no private benefit for the purchasing manager.
- A passive-waste public body benefits from Consip because it avoids costly search, incompetence, or inefficient procedures.
- An active-waste public body loses from Consip because it gives up the opportunity to obtain private benefits from outside suppliers.

The model implies:

$$\Pr(\text{Consip purchase}) \text{ is increasing in } \mu_i$$

and

$$\Pr(\text{Consip purchase}) \text{ is decreasing in } \beta_i.$$

Interpretation: This is the paper's core revealed-preference logic. If high-price public bodies move toward Consip, price differences are more likely to reflect passive waste. If high-price public bodies avoid Consip, price differences are more likely to reflect active waste.

5.3 Price fixed-effect regression

When Consip is not active, the authors estimate

$$\log p_{igt} = X'_{igt}\gamma_g + \rho_g \log Q_{igt} + \eta_g t + \theta_g + w_i + \varepsilon_{igt}.$$

- p_{igt} is the price paid by public body i for good g at time t.
- X_{igt} contains good-specific characteristics.
- Q_{igt} is quantity, allowing for bulk discounts.
- $\eta_g t$ allows good-specific time trends.
- θ_g are good fixed effects.
- w_i is the public-body fixed effect, interpreted as the public body's average open-market price level.

Interpretation: The estimated w_i is the empirical measure of how expensive a public body is when it must purchase outside Consip.

5.4 Reduced-form Consip-choice regression

When Consip is active, the authors estimate

$$C_{igt} = \alpha \widehat{w}_i + \eta_g t + \psi_g + v_{igt},$$

where $C_{igt} = 1$ if the public body buys from Consip and 0 otherwise.

- The coefficient α is the key reduced-form statistic.
- If $\alpha > 0$, high open-market-price public bodies are more likely to use Consip, which supports passive waste.
- If $\alpha < 0$, high open-market-price public bodies are less likely to use Consip, which supports active waste.
- The standard errors are clustered at the public-body-good level.

Interpretation: The paper finds $\alpha > 0$. This is the first major empirical indication that passive waste is more important than active waste in explaining price differences.

5.5 Structural decomposition

For quantification, the authors impose functional forms. The price function can be written conceptually as

$$f_{gt}(b_{igt}, \mu_i) = (1 + \mu_i + b_{igt}^2)\bar{p}_{gt}\varepsilon_{igt},$$

where $\bar{p}_{gt}$ is a reference price.

The structural price equation is

$$\log p_{igt} = \log \bar{p}_{gt} + \omega_i + \log \varepsilon_{igt},$$

with

$$\omega_i = \log(1 + \mu_i + \beta_i^2).$$

The switching equation implies

$$\Pr(\text{Consip}) = \Pr(-v_{igt} < \sigma_i + c_g),$$

with

$$\sigma_i = \mu_i - \beta_i^2.$$

- The price equation identifies total waste, $\mu_i + \beta_i^2$.
- The switching equation separates passive and active components because μ_i raises Consip use and β_i^2 lowers Consip use.
- The model requires a reference price. The paper uses three alternatives: the imputed Consip price, the imputed price paid by bottom-decile public bodies, and the minimum of the two.

Interpretation: The structural model converts the reduced-form sign into a quantitative decomposition of waste.

6 Identification and empirical design

6.1 What identifies active versus passive waste

- Without Consip, high prices can reflect either high active waste or high passive waste.
- With Consip, the two forms of waste imply opposite behavior.

- Passive-waste public bodies have a stronger incentive to use Consip because it helps them avoid inefficient procurement.
- Active-waste public bodies have a weaker incentive to use Consip because it removes the possibility of private benefits.
- Therefore, the covariance between open-market prices and Consip use identifies whether cross-public-body price differences are mostly active or passive.

Interpretation: The identifying variation is not merely the level of Consip prices. It is the observed choice to use or avoid Consip conditional on being allowed to use it.

6.2 Key assumptions

- Consip must treat public bodies symmetrically by offering the same price to all for a given good agreement.
- Buying from Consip should not itself generate the same type of private benefits as buying outside Consip.
- The timing of Consip agreements across goods should be plausibly unrelated to individual public bodies' active or passive waste parameters.
- Public bodies should not strategically time purchases in ways that are correlated with active or passive waste.
- For the reduced-form test, Consip does not have to be perfectly efficient or free of waste; it only needs to offer a common purchase option.
- For the structural decomposition, the reference-price assumptions are stronger and determine the level of estimated waste.

Interpretation: The reduced-form evidence is cleaner than the structural level estimates. The structural estimates add magnitudes but require stronger assumptions about the efficient reference price.

6.3 Alternative explanations

- **Endogenous monitoring.** If prosecutors target corrupt bodies and the Consip decision helps them audit, corrupt bodies might buy from Consip to hide. This could make active waste look like passive waste.
- The authors argue that this is unlikely in practice because not buying from Consip is not itself evidence of a crime, and Italian corruption investigations tend to rely on evidence of payments rather than statistical price comparisons.
- The compulsory regime provides a test. If fear of auditing explains the result, the high-price-to-Consip relationship should be stronger when Consip use is legally compulsory. The interaction is not significant and, if anything, weaker.
- **Active waste through Consip.** If managers could receive bribes when using Consip, corrupt managers might also choose Consip. The authors argue this is unlikely because Consip suppliers are large national or international firms and local purchasing managers order directly from them.

- **Quality differences.** The paper controls for detailed product characteristics and discusses evidence that Consip purchases do not simply involve lower-quality goods.

Interpretation: The alternative theories matter because they could reverse the paper's interpretation. The paper's institutional discussion and robustness tests are designed to show that the positive α is not simply hidden corruption behavior.

7 Tables and figures

7.1 Figures 1 and 2: theory and price dispersion

Read:

- Figure 1 shows the model's central prediction. If price differences are due to passive waste, high open-market prices should predict a higher probability of buying from Consip. If price differences are due to active waste, high open-market prices should predict a lower probability of buying from Consip.
- Figure 2 shows the distribution of public-body average prices for goods not purchased from Consip. The tenth percentile is 0.79 and the ninetieth percentile is 1.23, so the ninetieth-percentile body pays about 55 percent more than the tenth-percentile body.

Economic message: Figure 1 explains the identification; Figure 2 shows why the question matters economically. Price dispersion is large enough that diagnosing the type of waste is not a minor accounting exercise.

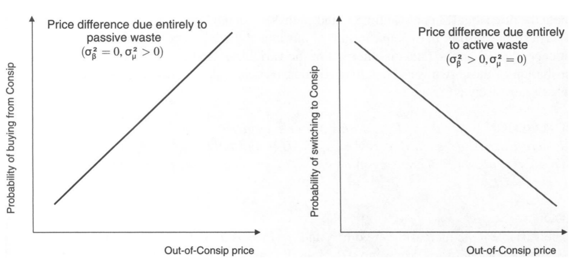

FIGURE 1. MODEL PREDICTION

(a) Figure 1: Model prediction.

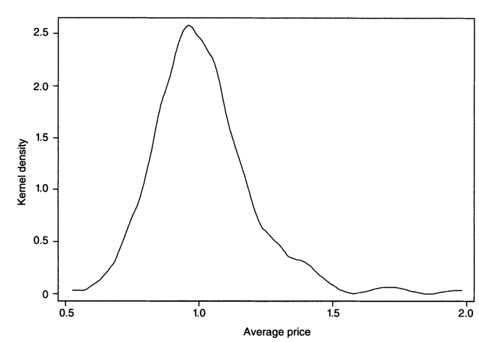

FIGURE 2. AVERAGE PRICES OF GOODS NOT PURCHASED FROM CONSIP

Percentile	1	10	50	90	99
Average price	0.67	0.79	1	1.23	1.68

Notes: The average price for out-of-Consip purchases is estimated for each PB as the exponent of PB *i*'s fixed effect in the regression of log price on: goods fixed effects, good-specific trends, good-specific quantities, and good-specific characteristics, using the sample of purchases made when a Consip agreement was not active. The average out-of-Consip price for the PB at the tenth percentile is 0.79, and for the PB at the ninetieth percentile is 1.23.

(b) Figure 2: Average prices of goods not purchased from Consip.

7.2 Tables 1A and 1B: sample description

Read:

- Table 1A describes public bodies by governance class. The sample includes ministries, social security administrations, regional councils, province and town councils, health centers, mountain village councils, universities, and other bodies.

- Central bodies are more likely to buy from Consip when a Consip agreement is active. For example, ministries and government bodies purchase through Consip for 62 percent of purchases made when agreements are active.
- Table 1B describes the 21 goods. The sample covers 6,068 purchases and includes car rentals, computers, office furniture, phone contracts, fuel, lunch vouchers, paper, software, printers, servers, cars, and fax machines.
- The goods differ sharply in Consip coverage and Consip take-up. Some goods, such as refuse bins, servers, and car purchases, have no Consip agreement in the sample period.

Economic message: The data are broad enough to compare many public bodies and goods, but narrow enough to focus on standardized products for which price comparisons are meaningful.

TABLE 1A—SAMPLE DESCRIPTION
(*Public bodies sample)*

Governance class	Number of PBs	Total expenditure by sample PBs in 2000 (€ million)	Total expenditure by sample PBs over total expenditure by all PBs in 2000 (€ million)	Average number of goods purchased	Percentage of total purchases made when a Consip agreement is active	Percentage of Consip purchases made when a Consip agreement is active
Napoleonic bodies						
Ministries and government	12	13,368	0.92	12.2	59	62
Social security administration	3	1,952.90	0.78	10.5	61	57
Local governments						
Regional councils	12	1,683.90	0.61	10.6	51	26
Province and town councils	70	4,162.20	0.21	11.9	51	39
Semi-autonomous bodies						
Health centres	81	6,894.20	0.48	11.8	56	35
Mountain village councils	11	34.2	0.13	10.5	54	33
Universities	13	354.5	0.43	12	53	34
Other	6	462.3	0.29	11.8	44	45

Notes: The "other" category includes: the National Statistical Institute (ISTAT), the Institute for International Trade (ICE), the Higher Institute of Health (ISS), the National Research Institute (CNR), a Veterinary Research Center, and a Regional Research Institute. Total expenditure by sample PBs equals yearly expenditure for goods and services summed over all sample PBs in a given class. Total expenditure by all PBs equals yearly expenditure for goods and services summed over all the PBs belonging to that institutional class.

Source: ISTAT.

(a) Table 1A: Public bodies sample.

TABLE 1B—SAMPLE DESCRIPTION
(*Goods sample)*

Good type	Observations (1)	Average price (2)	Average quantity per order (3)	Percentage of days when a Consip agreement is active (4)	Percentage of Consip purchases when an agreement is active (5)
Car rental	160	399.5	4.81	53	68
		(208.6)	(9.58)		
Photocopier rental	466	510.69	13.06	58	64
		(844.52)	(30.18)		
Laptop	775	1,219.7	6.5	45	35
		(458.52)	(30.1)		
Desktop	648	992.5	16.0	39	47
		(587.5)	(62.84)		
Office desk	245	232.1	11.9	10	11
		(171.9)	(26.02)		
Office chair	280	96.6	30.4	25	5
		(52.7)	(86.2)		
Landline contracts	143	1.89	125,272	50	94
		(0.74)	(292,636)		
Projector	191	1,438.0	1.82	13	44
		(647.3)	(2.44)		
Local network: switch	215	138.7	164.4	33	7
		(269.9)	(298.5)		
Local network: cable	102	3.33	8,631.1	33	26
		(4.71)	(3,245.3)		
Motor oil	23	5.19	681.34	41	0
		(2.01)	(1,155.1)		
Heating diesel	248	3.85	293,583	50	30
		(13.81)	(504,625)		
Lunch vouchers	231	70.04	665,895	79	52
		(4.57)	(1,418,723)		
Refuse bins	63	152.63	290.76	0	0
		(184.94)	(768.58)		
Paper	755	2.40	6,546.5	32	9
		(0.922)	(2,2626.2)		
Mobile phone contracts	183	0.041	1,244,620	57	59
		(0.102)	(5,011,294)		
MS Office software	155	233.2	151.1	56	20
		(91.5)	(483.1)		
Printer	294	483.95	22.6	43	47
		(576.7)	(96.9)		
Server	297	5,967.5	3.45	0	0
		(6,772.6)	(9.24)		
Car purchases	345	10,710.3	4.02	0	0
		(6,112.7)	(11.23)		
Fax	249	338.16	6.89	45	41
		(158.85)	(18.02)		
Total	6,068				

Notes: For goods purchases, price equals the cost of one unit. Motor oil and heating diesel are measured in liters, cables are measured in meters. For goods rentals, price equals the monthly rent for one unit of the good. For landline contracts, price equals the per-minute charge for national calls. For mobile contracts, price equals the per-minute charge for calls to landlines. Quantity equals the number of items in a single purchase, except heating diesel and motor oil, where quantity is measured in liters, cables, where quantity is measured in meters, and landline, mobile, and lunch vouchers, where quantity is measured as total yearly outlay. Column 4 reports the number of days during which an agreement was active over the total number of days in our sample. During our sample, Consip did not make agreements for refuse bins, car purchases, and servers. Column 5 reports the number of purchases from the Consip catalogue divided by the total

(b) Table 1B: Goods sample.

7.3 Tables 2 and 3: governance correlates and reduced-form test

Read:

- Table 2 regresses out-of-Consip average prices on geography, size, social capital, and governance type.

- Once governance type is included, geography, size, and social-capital variables are not robustly significant.
- Relative to universities, ministries and government bodies pay about 39–40 percent more, social security institutions about 22 percent more, regional councils about 21 percent more, and province/town councils about 13 percent more in the richest specification.
- Table 3 is the key reduced-form test. The coefficient on out-of-Consip price is positive and statistically significant across specifications.
- In column 3, the coefficient is 0.219. Moving from the tenth to the ninetieth percentile of out-of-Consip prices raises the probability of buying from Consip by about 9.7 percentage points.
- The compulsory-regime dummy is positive, but adding it does not change the key coefficient. The interaction with compulsory regime is negative and not significant.

Economic message: Table 2 shows that governance is central to price dispersion. Table 3 then shows that high-price bodies are more likely to use Consip, which is the core evidence for passive waste.

TABLE 2—PRICES AND PB CHARACTERISTICS
(Dependent variable is the average price for out-of-Consip purchases: Linear model–standard errors in parentheses)

	(1)	(2)	(3)	(4)
Geography (Omitted: North)				
South-oc	0.019			−0.036
	(0.050)			(0.095)
South	−0.005			−0.062
	(0.034)			(0.059)
Center	0.082**			0.008
	(0.034)			(0.036)
Size				
Log expenditure	0.024**			−0.010
	(0.011)			(0.018)
Social capital variables				
Donations		−2.43**		−1.23
		(0.856)		(0.758)
Referenda voter turnout		0.172		−0.145
		(0.268)		(0.482)
Share of people who trust		0.024		0.147
		(0.143)		(0.158)
Governance types (Category omitted: university)				
Napoleonic bodies				
Ministries and government			0.397***	0.394***
			(0.103)	(0.138)
Social security			0.229***	0.224***
			(0.036)	(0.076)
Local bodies				
Regional councils			0.184***	0.207***
			(0.054)	(0.069)
Province and town councils			0.109***	0.126***
			(0.029)	(0.034)
Semi-autonomous bodies				
Health centers			0.038	0.063*
			(0.028)	(0.037)
Mountain village councils			0.030	0.065
			(0.065)	(0.074)
Adjusted R^2	0.0689	0.0592	0.2320	0.2515
Observations	202	189	202	189

Notes: The omitted category for the type variable is "universities." The omitted category for the geographical variable is "North." South-oc identifies the southern regions with high prevalence of organized crime (Campania, Puglia, Calabria, and Sicilia). Six PBs that do not belong to any of the three governance classes are excluded from the sample. Donation is the number of blood bags (each bag contains 16 ouncesof blood) per million inhabitants in province collected by AVIS, the Italian association of blood donors, in 1995 among its members. Referenda voter turnout covers participation in all referenda between 1946 and 1987 averaged through time at the province level. Share of people who trust is measured at the province level from the World Value Survey for Italy 1990 and 1999. Sample size falls because these variables are not available for eight provinces. More details on the construction of the social capital variables can be found in the appendix of Guiso et al. (2004), who kindly provided the data.
*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

(a) Table 2: Prices and public-body characteristics.

TABLE 3—BUYING FROM CONSIP AS A FUNCTION OF OUT-OF-CONSIP PRICES
(Dependent variable = 1 if good purchased via Consip: Linear probability model–standard errors clustered by PB-good type in parentheses)

	Baseline (1)	Good FE (2)	Trends (3)	Different regimes (4)	Regimes interaction (5)
Out-of-Consip price	0.228***	0.232***	0.219***	0.193***	0.253***
	(0.078)	(0.063)	(0.059)	(0.057)	(0.083)
Compulsory regime (=1 if yes)				0.240***	0.240***
				(0.028)	(0.028)
Out-of-Consip price × compulsory regime					−0.107
					(0.105)
Good FE	No	Yes	Yes	Yes	Yes
Good specific trends	No	No	Yes	Yes	Yes
R^2	0.0060	0.2429	0.2753	0.3037	0.3040
Observations	3,122	3,122	3,122	3,122	3,122

Notes: Out-of-Consip price is estimated as PB *i*'s fixed effect in the regression of log price on goods fixed effects, good specific trends, good specific quantities, and good specific characteristics, using the sample of purchases made when a Consip agreement was not active. The compulsory regime variable equals one when PBs were required to buy from Consip if there was an active agreement for an equivalent good, unless they could claim that the good's characteristics were not suited to their needs. This regime applied to PBs belonging to the Central Public Administration between 2000 and 2002 and to all PBs in 2003.
*** Indicates significance at 1 percent.
** Indicates significance at 5 percent.
* Indicates significance at 10 percent.

(b) Table 3: Buying from Consip as a function of out-of-Consip prices.

7.4 Table 4 and Figure 3: structural active/passive decomposition

Read:

- Table 4 reports the structural estimates under three reference-price assumptions: Consip prices, bottom-decile public-body prices, and the minimum of the two.
- The average public body pays 43 percent more than the Consip reference price and 21 percent more than the bottom-decile reference price.
- Across assumptions, the estimated passive share of waste is very stable: about 82–83 percent.
- Passive waste accounts for more than half of total waste for about 83–84 percent of public bodies.
- Active waste is positive for a minority of bodies: about 19–29 percent depending on the reference-price assumption.
- Figure 3 plots public-body-level passive waste against active waste. The scatter shows a wider range of passive waste and little obvious correlation between active and passive waste.

Economic message: The structural estimates turn the reduced-form sign into magnitudes. The main economic claim is not that corruption is zero, but that passive waste is the dominant component of price overpayment for these standardized goods.

TABLE 4—ESTIMATES OF PASSIVE AND ACTIVE WASTE

Reference price	Average waste $(1 + \mu + b^2)$ (1)	Average passive waste (μ) (2)	Average active waste (b^2) (3)	Share of passive waste $(\mu/(\mu + b))$ (4)	Share of PBs for which $(\mu/(\mu + b)) > 0.5$ (5)	Share of PBs for which $b > 0$ (6)
Consip	1.43	0.320	0.093	0.833	0.838	0.226
Bottom decile PBs	1.21	0.148	0.053	0.822	0.828	0.192
Minimum (Consip, bottom decile)	1.56	0.425	0.113	0.829	0.843	0.288

Notes: Passive and active waste are estimated from the price equation (6) and the selection equation (7), as explained in Section VC. The reference prices in rows 1, 2, and 3 are, respectively, the estimated Consip price for a good with the same characteristics, the estimated price paid by PBs in the bottom decile of the distribution of average out-of-Consip prices for a good with the same characteristics, and the lesser of the two.

(a) Table 4: Estimates of passive and active waste.

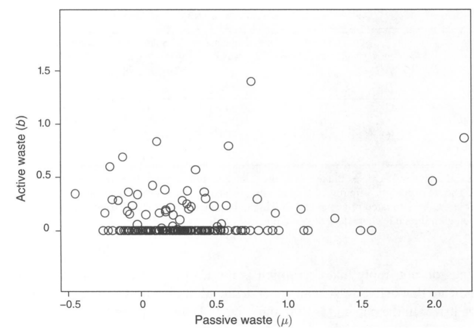

FIGURE 3. ACTIVE AND PASSIVE WASTE

Notes: Each point represents a different PB. For each PB, passive and active waste are estimated from the price equation (6) and the selection equation (7) as explained in Section VC, under the assumption that the reference price is the estimated Consip price for a good with the same characteristics.

(b) Figure 3: Active and passive waste.

8 Short summary

- The paper studies why Italian public bodies pay different prices for standardized goods.
- It separates active waste, which benefits the decision maker, from passive waste, which does not.
- Consip, a centralized procurement agency, creates a purchase option that helps distinguish the two forms of waste.
- High open-market-price public bodies are more likely to use Consip when it is available.
- This positive relationship is evidence that price dispersion is mainly due to passive waste.
- Governance structure predicts procurement prices: central Napoleonic bodies perform worst; semi-autonomous bodies perform best.
- The structural model estimates that at least 82 percent of waste is passive.

- Passive and active waste are not strongly negatively related; more autonomous bodies have lower passive waste without higher active waste.
- Consip can generate large savings, but a mandatory centralized system could reduce flexibility and change Consip's incentives.
- The paper's broader message is that corruption is not the only, and may not be the largest, source of government inefficiency.

9 Conclusion

9.1 Core conclusion

Bandiera, Prat, and Valletti show that excessive procurement prices paid by Italian public bodies for standardized goods are mainly due to passive waste rather than active waste. Public bodies differ greatly in the prices they pay, but the bodies that pay more on the open market are more likely to switch to Consip when they can. In the model, this pattern is the signature of passive waste: inefficient bodies value a centralized low-price purchase option.

9.2 Economic intuition

The intuition is simple. A corrupt or rent-seeking purchasing manager does not necessarily want a centralized procurement option because it removes discretion and private benefits. An inefficient purchasing manager does want such an option because it lowers the cost of finding and executing a good purchase. Thus, the Consip decision reveals the nature of the hidden procurement problem. High price plus high Consip take-up points to passive waste; high price plus Consip avoidance would point to active waste.

9.3 Incremental contribution revisited

The paper advances the government-efficiency literature in three main ways. First, it offers a clean conceptual distinction between waste that benefits the official and waste that does not. Second, it designs an empirical test based on revealed preference rather than opinion surveys or corruption perceptions. Third, it quantifies the active and passive components using a structural model. This moves the discussion from "government overpays" to "why government overpays."

9.4 Policy implications

The policy implications differ sharply by type of waste. If active waste dominates, stronger rules, audits, and monitoring are natural responses. If passive waste dominates, policy should improve procurement capacity, information, incentives, and managerial discretion. Consip-style centralized purchasing can help by giving public bodies an easy benchmark and purchase channel. However, making Consip compulsory could impose specification mismatches and weaken Consip's incentive to find attractive prices if it no longer has to compete for demand.

9.5 Limitations

The analysis focuses on standardized goods, which are easier to compare than unique procurement projects such as roads, software systems, or complex public works. Active waste may be more important in those settings because quality, specifications, and local supplier relationships are harder to observe. The structural estimates also depend on assumptions about the efficient reference price. Finally, governance differences may partly reflect sorting of managers across public bodies, not only institutional rules.

9.6 Takeaway

The enduring takeaway is that corruption is only one form of public-sector inefficiency. In this setting, the larger problem is passive waste: public bodies overpay because they lack the skills, incentives, or organizational structure to buy well. This distinction matters because the wrong diagnosis can produce the wrong policy. More rules may reduce corruption, but if the main problem is passive inefficiency, better procurement support and better governance may matter more.